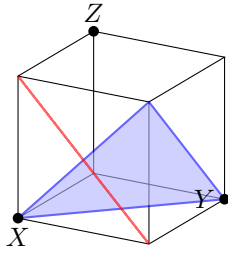


## 1 Exercise 1

consider the cube  $C = (x, y, z) \in \mathbb{R}^3 | 0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$  in  $\mathbb{R}^3$ . Find the angle between the plane  $P_1$  containing the 3 points:  $(0, 1, 0), (1, 0, 0), (1, 1, 1)$  and the plane  $P_2$  containing the 3 points:  $(0, 0, 0), (1, 1, 0), (1, 0, 1)$ .



To find the angle between the two planes, we must firstly find the normal vector that the individual planes define and then find the angle between those two.

$$\vec{P_{1X}} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{P_{1Y}} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \vec{P_{1Z}} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Vectors for  $P_1$